

From Topic to Talk in One Pipeline

A 45-minute practical with six AI research tools

LOURIM, AI for Research Seminar Series, Part 1

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What you will do today. You received a research topic on a slip of paper, randomly assigned by a peer. By the end of this session you will have *started* a pipeline that ends, at home, with a short paper draft and a slide deck on that topic, which you will email to a researcher working in the field. The six tools below each take care of one step.

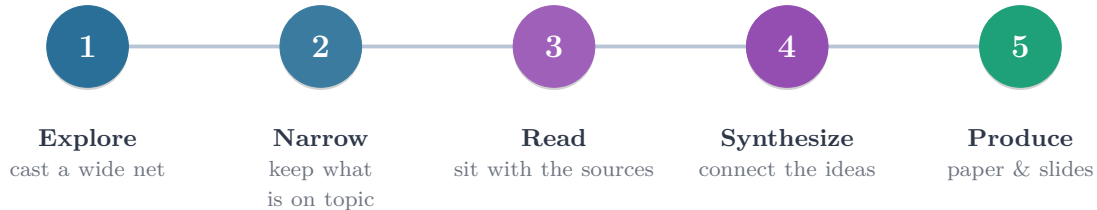
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1 Overview

1.1 From topic to talk: the method

Before we name any tool, here is the recipe behind doing good research, with or without AI. Five steps, in order:



Explore. You start from a topic, not a question. Look broadly at what already exists, even outside your own field, so you do not miss the literature that matters.

Narrow. Filter the long initial list into a small, trustworthy set you can actually engage with. Drop the off-topic and the weak.

Read. Spend real time with each source. Note the question it asks, the method it uses, what it finds, and where it is fragile.

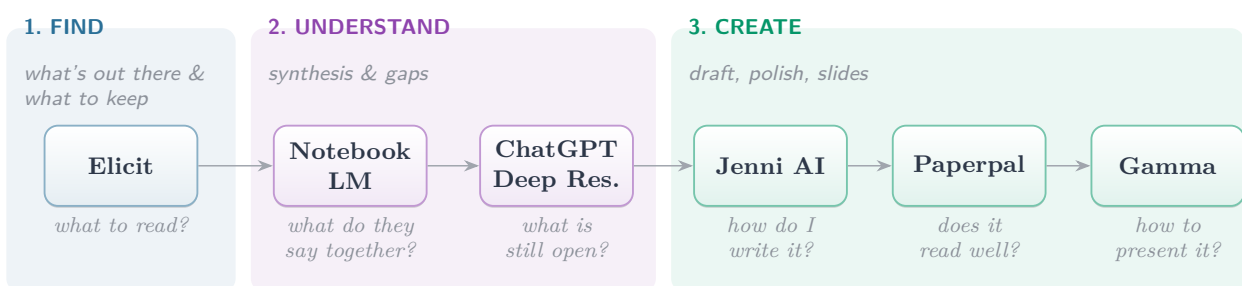
Synthesize. Step out of individual papers and look at the picture they make together: what is the consensus, where do they disagree, what is missing?

Produce. Turn that synthesis into something other people can engage with: a written argument, and a presentation that does it justice.

This recipe is the work of any researcher. The six AI tools below are nothing more than *assistants* for each step. They speed up the recipe, they do not replace it.

1.2 The six tools

Each tool takes care of one step of the recipe and answers one specific question:



1.3 Time budget

The 45-minute session is a **kickoff**, not the whole journey. The plan below is a guideline. The real instruction is simple:

Tip

Go as far as you can in 45 minutes. Some of you will finish Elicit only, some will reach NotebookLM, and that is fine. The rest is guided homework you finish over the following days. What matters is that you start the pipeline today and that you keep moving forward at your

own pace.

Phase	What happens	Suggested time
Intro	Recap pipeline; check accounts; pick your topic phrasing	5 min
§ 3 Elicit	Search, filter, build the table, download 3 PDFs	20 min
§ 4 NotebookLM (start)	Upload the papers, run the synthesis questions	15 min
Wrap-up	Brief on what to do at home	5 min
Homework	At your own pace, around 3 to 5 h spread over a week	

If you fall behind in the room, do *not* skip steps to catch up. Stop where you are, take a clean note of where you stopped, and pick up from there at home. Skipping steps is what produces a paper you cannot defend.

1.4 Golden rule

Tip

You are the editor; the AI drafts. Every output of every tool is a starting point you read, prune, and correct. If you cannot defend a sentence to a colleague, delete it.

2 Before you start

1. **Topic in hand.** The slip of paper a peer gave you. If it feels too broad (e.g. “leadership”), narrow it to one question (e.g. “how does remote work change leadership in teams?”).
2. **A browser open** with you signed in to the six tools. The free versions are enough; sign-up links are in Appendix A.
3. **A folder for this project**, somewhere you will find it again, on your computer, in Dropbox, in Google Drive, wherever you usually work. Inside it, make four sub-folders: *Papers*, *Notes*, *Draft*, *Slides*. You will save things there as you go.
4. **A document to keep your prompts.** Open a Word document (or a Google Doc, or Notes) and call it “My prompts”. Each time you ask one of the tools a question that worked well, paste the question in. Later, this lets you remember what you did and explain it if asked.
5. **Do not upload anything sensitive or unpublished**, your own work, your colleagues’ work, or interviews you collected. Stick to published papers and to notes you are happy to share.

3 Elicit: search and screen

Goal. Find the 3 papers most relevant to your topic, and a quick read of the field around them.

Input. Just your topic in a few words.

Output. A comparison table you can read in Elicit (research question, main finding, year), plus 3 downloaded PDFs in your *Papers* folder.

Sign up

Go to elicit.com and click “Sign up”. The free version is enough for this exercise.

Steps

1. In Elicit, choose the *Find papers* workflow. Type your topic phrased as a research question (for example, “how does remote work change leadership in teams?”). Elicit will return a ranked list of papers.
2. Apply three filters before you look at any abstract:
 - *Has PDF available*
 - *Q1 journals only*
 - *Published in the last 6 years*
3. Add three columns to the table: *Research question*, *Main finding*, *Year*.
4. Click “Run” so Elicit fills the cells. Read them. If something looks wrong, it probably is. Elicit sometimes summarizes too confidently.
5. Exporting the table is not available on the free plan, so pick the 3 papers that look most relevant to you, google their titles, and download the PDFs into your *Papers* folder.

Watch out

Elicit can invent. The cells are summaries written by an AI, not quotes from the paper. Before you put a number, a sample size, or a finding into your own paper, open the PDF and check it with your own eyes.

Time budget: 20 min.

4 NotebookLM: synthesize your corpus

Goal. Understand, as a whole, what your 3 papers are saying together.

Input. The 3 PDFs you downloaded.

Output. A set of notes about the main themes, agreements, disagreements, and open questions.

Sign up

Go to notebooklm.google.com and sign in with a Google account. It is free.

Steps

1. Click *New notebook* and give it the name of your topic.
2. Upload the 3 PDFs you downloaded as “sources”.
3. On the right side there is a chat box. Paste the four questions from the *Copy-paste prompt* box just below, one by one. After each answer, click *Save to note* so it lands in your Notes panel.
4. Once all four answers are saved, open the note in NotebookLM and use the *Export to Google Docs* option. Move the resulting Doc into your *Notes* folder.

Copy-paste prompt

1. What are the 4 to 6 main themes that recur across these sources?
2. Where do the sources agree, and where do they disagree? Cite the sources.
3. Which definitions of the central concepts vary across the sources, and how?

4. What questions do these papers leave unanswered?

Watch out

A citation does not mean “correct”. NotebookLM points you to the file it used, but its summary of that file can still be wrong. Click each little citation number to jump to the source and check.

Time budget: 15 min in session to upload and run prompts; finish curation at home.

5 ChatGPT Deep Research: find the gaps

Goal. Step back from your papers and ask: what are the open questions in this field?

Input. The notes you saved from NotebookLM.

Output. A short “insights” document you will use to write the paper.

Sign up

Go to chatgpt.com. Deep Research is only available on the paid plan. If you do not have it, the same prompt also works in regular ChatGPT, or you can use *Undermind* (link in the troubleshooting appendix).

Steps

1. In ChatGPT, open the tools menu (the small icons next to the message box) and choose *Deep Research*.
2. Copy the prompt below into the message box. Replace <TOPIC> with your topic, and paste your NotebookLM notes underneath where it says “NOTES”.
3. Send it. Deep Research takes 5 to 15 minutes; this is normal. You can leave the tab open and do something else.
4. When the report comes back, read it slowly. For each claim you want to keep, click the source link. If the link is broken, or if the source does not actually say what is claimed, drop the claim.
5. Save the cleaned report in your *Notes* folder under the name “Insights”.

Copy-paste prompt

You are a research assistant for a literature-review draft on <TOPIC>.
Below are my synthesized notes from a small curated corpus. Using these notes plus the broader literature, produce:

1. The 3 to 5 most consequential research gaps.
2. The main methodological challenges in the field.
3. A side-by-side comparison of the dominant theoretical perspectives.
4. Two or three concrete, testable directions for future work.

For every claim, cite at least one source with a working link. Flag uncertainty explicitly.

=== NOTES ===

<paste NotebookLM notes here>

Watch out

Made-up citations. Even Deep Research sometimes invents references that look real. Open every link before you trust a claim. If the link is dead, or the page does not say what was

claimed, throw the claim out.

Time budget: 15 min active plus waiting time. Homework.

6 Jenni AI: draft the paper

Goal. Write a first draft of a short paper (about 1 500 to 2 000 words).

Input. Your NotebookLM notes and your ChatGPT insights.

Output. A first draft of a mini-paper, organized in sections.

Sign up

Go to jenni.ai. The free version has a small daily word limit, enough to write one short paper.

Steps

1. Click “New document” and use your topic as the title.
2. In the side panel, paste your NotebookLM notes and your ChatGPT insights so Jenni can use them as background.
3. Type the section titles, one under the other: *Introduction, Background, What the Literature Says, Open Questions, Where to Go Next, Conclusion*.
4. Under each section title, write a single sentence saying what you want this part to argue. Then let Jenni’s autocomplete suggest the next sentence. Accept it, change it, or reject it, one suggestion at a time.
5. As you write, add citations from the papers you kept. Jenni offers a button for this.
6. Save the draft as a Word document into your *Draft* folder, under the name “Paper, version 1”.

Tip

Always write the first sentence of a paragraph yourself. It steers Jenni and cuts down a lot on invented content.

Watch out

Wrong citations. Jenni may add a citation that looks perfect but does not actually support the sentence. Open the paper and check before you keep it.

Time budget: 60 to 90 min. Homework.

7 Paperpal: polish the language

Goal. Improve the tone, the clarity, and the grammar so it reads like a proper academic text.

Input. Your draft from Jenni.

Output. A final, polished version of the paper, saved as Word and as PDF.

Sign up

Go to paperpal.com. The free version is enough for one paper. If you use Microsoft Word, installing the Paperpal Word add-in is the easiest option.

Steps

1. Upload your Jenni draft to Paperpal, or open it in Word and turn on the Paperpal add-in.
2. Run the *Language quality check*. Look at the suggestions one by one. Accept the ones that improve clarity. Reject any suggestion that changes the meaning.
3. Run the *Pre-submission check* (or the *Subject-area check*, if it asks).
4. Use the *Paraphrase* button only for one awkward sentence at a time. Do not paraphrase a whole paragraph at once.
5. Save the polished version twice in your *Draft* folder: once as a Word document, once as a PDF. Call them “Paper, final”.

Watch out

Polishing can change what you said. Paperpal happily rewrites a careful “may” into a confident “does”. Re-read each suggestion and reject it when it overstates your claim.

Time budget: 30 min. Homework.

8 Gamma: generate the slide deck

Goal. Turn the paper into a slide deck of 7 to 10 slides.

Input. Your final paper (the PDF, or its text).

Output. A slide deck saved as a PDF.

Sign up

Go to gamma.app. The free version has enough credits for one deck.

Steps

1. Click *Create new*, then *Generate*, then choose “from text” (or “Text transform”).
2. Paste the text of your paper (the whole thing, or a one-page summary).
3. Set the options: about 8 to 10 slides, tone “academic”, audience “researchers”.
4. Click *Generate*. After a moment, the deck opens.
5. Edit the first slide so it shows your name, your topic, the date, and LOURIM. Add or remove slides so the order matches the sections of the paper.
6. If Gamma added an image you cannot defend, replace it or delete it. When in doubt, no image is better than the wrong image.
7. Click *Export* and save as PDF in your *Slides* folder, under the name “Slides”.

Tip

You can do Gamma at the same time as Paperpal: while Paperpal is checking your draft, start Gamma generating in another tab.

Watch out

Generated charts are decoration, not data. If Gamma adds a chart, it is invented from the words of your paper, not from real numbers. Never present it as if it were a real measurement.

Time budget: 30 min. Homework.

9 Send to the researcher

The last step, about a week after the session, is to send your work to a researcher who works on your topic.

Choose the person

Pick the researcher you (or the organizer) identified for your topic. The simplest choice is the author who comes up most often in the papers you kept, and whose university email address is easy to find online. One person is enough.

What to attach

- Your final paper (the PDF).
- Your slide deck (the PDF).

Email template

Copy-paste prompt

Subject: Student exploration on <TOPIC>, LOURIM AI seminar (your feedback?)

Dear Prof. <NAME>,

As part of an AI for research seminar at LOURIM (UCLouvain), I produced a short literature synthesis on <TOPIC>, drawing notably on your work on <SPECIFIC PAPER>.

The paper draft and a short slide deck are attached. The exercise was time-boxed and AI-assisted, so it is exploratory rather than a finished review. I would value any feedback you might have on the framing, gaps, or proposed direction.

With thanks for your time,
<YOUR NAME>
LOURIM, UCLouvain

Watch out

Disclose AI use. The email already says “AI-assisted”; keep that line. Researchers appreciate transparency far more than they mind the use of tools.

10 Risks: a six-line reminder

Hallucinations	Check every fact, number, and citation against the original source.
Bias	These tools reflect what they were trained on. Be aware whose voices may be missing.
Confidentiality	Never upload unpublished data, sensitive interviews, or internal LOURIM documents.
Traceability	Keep your “My prompts” document up to date. You will be glad later.
Scientific integrity	You have to be able to defend each argument as your own.
Responsibility	It is your name on the paper, not the tool’s.

A Accounts and free tiers

Tool	Sign-up	Free-tier note
Elicit	elicit.com	Free monthly credits, enough for one screening.
NotebookLM	notebooklm.google.com	Free with a Google account.
ChatGPT Deep Res.	chatgpt.com	Paid plan; fallback: regular ChatGPT or Undermind.
Jenni AI	jenni.ai	Free daily word cap; fits one mini-paper.
Paperpal	paperpal.com	Free checks per month plus Word add-in.
Gamma	gamma.app	Free credits cover one deck.

B Troubleshooting

- **Elicit returns very few papers after filtering.** Loosen one filter at a time: try the last 8 years instead of 6, or drop the Q1 restriction. Re-check that your topic is phrased as a question, not a single keyword.
- **Elicit ran out of free credits.** Run the extraction only on the 3 papers you plan to keep, not on the full ranked list.
- **NotebookLM refuses to upload a PDF.** The file is probably too big or it is a scanned image. Split the PDF, or use a free tool like Adobe to turn the scan into real text.
- **No access to Deep Research.** Use *Undermind* (undermind.ai) instead, or paste the same prompt into the regular ChatGPT with the web search option turned on.
- **Jenni adds citations you do not recognize.** In Jenni’s settings, turn off the option that lets it search the web for citations. Keep only the papers you uploaded.
- **Paperpal changes the meaning.** Reject the suggestion and rewrite the sentence by hand. Never click “Accept all” on the whole document.
- **Gamma exports an ugly PDF.** Use the built-in *Export to PDF* button, not the browser’s “Print” function.

LOURIM, AI for Research Seminar Series, Part 1. Source: github.com/cvanderkerckh/lourim-ai-seminar-series.